Online Appendix to “Dynamic Batch Learning for Online Decision-Making in Smooth Contextual Bandits”

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This Supplementary Material provides the complete mathematical derivations and technical details for the lemmas and auxiliary results presented in the main paper. It consists of the following sections:

- Section 1 presents the detailed proofs for the primary lemmas that appear in the main manuscript.
- Section 2 contains the comprehensive proofs and technical derivations for the auxiliary lemmas previously introduced in the supplementary material.

1. Proofs of Lemmas in the Main Manuscript

Proof of Lemma 1 We prove this lemma by induction. When $m = 1$, it is trivially true that $Q_a \subseteq \mathcal{R}_{1,a} = \mathcal{X}$ and $\max_{a\in\mathcal{A}} f_a(\mathbf{x}_t) - \min_{a\in\mathcal{A}} f_a(\mathbf{x}_t) \leq 2$. Assume that the lemma holds for $m \leq m_0$, we need to proof that it also holds for $m_0 + 1$. Under the event $\overline{\mathcal{G}}_{m_0} \cap \overline{\mathcal{M}}_{m_0}$, if $\mathbf{x}_t \in Q_a \subseteq R_{m_0,a}$, we have Q_a is (c_0, r_0)-regular at $\mathbf{x}_t$ and according to Lemma EC.1, $R_{m_0,a}$ is also (c_0, r_0)-regular at $\mathbf{x}_t$. Assume that $\mathbf{x}_t \in D_{m_0,a}$, then according to the definition of irregular set and Lemma EC.3, Lemma EC.4, we can derive that $R_{m_0,a}$ is not (c_0, r_0)-regular at $\mathbf{x}_t \in D_{m_0,a}$. Therefore, we have $Q_a \subseteq R_{m_0,a} \cap D^C_{m_0,a}$. Furthermore, if $\mathbf{x}_t \in Q_a$, $f_a(\mathbf{x}_t) \geq \max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)} f_a(\mathbf{x}_t)$ and considering the event $\mathcal{G}_{m_0}$, $|\hat{f}_{m_0,a}(\mathbf{x}_t) - f_a(\mathbf{x}_t)| \leq \frac{\epsilon_{m_0}}{2}$ we can derive that

$$\hat{f}_{m_0,a}(\mathbf{x}_t) + \frac{\epsilon_{m_0}}{2} \geq f_a(\mathbf{x}_t) \geq \max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)} f_a(\mathbf{x}_t) \geq \max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)} \hat{f}_{m_0,a}(\mathbf{x}_t) - \frac{\epsilon_{m_0}}{2}.$$

This is equal to $\hat{f}_{m_0,a}(\mathbf{x}_t) \geq \max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)} \hat{f}_{m_0,a}(\mathbf{x}_t) - \epsilon_{m_0}$. according to the definition of active arm set $\mathcal{A}_m(\mathbf{x}_t)$ in Algorithm 1, we have $a \in \mathcal{A}_{m_0+1}(\mathbf{x}_t)$ and that is $\mathbf{x}_t \in R_{m_0+1,a}$. Thus we have $\mathcal{Q}_a \subseteq \mathcal{R}_{m_0+1,a}$.

We then proof that for m_0+1, $\max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} f_a(\mathbf{x}_t) - \min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} f_a(\mathbf{x}_t) \leq 2\epsilon_{m_0}$. As $\mathcal{A}_{m_0+1}(\mathbf{x}_t) \subseteq \mathcal{A}_{m_0}(\mathbf{x}_t)$ and $\mathcal{Q}_a \subseteq \mathcal{R}_{m_0+1,a}$, then $a^*_t \in \mathcal{A}_{m+1}(\mathbf{x}_t)$ and under event $\mathcal{G}_{m_0}$, we have

$$\begin{aligned}
\max_{a\in\mathcal{A}} f_a(\mathbf{x}_t) - \min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} f_a(\mathbf{x}_t) &= \max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} f_a(\mathbf{x}_t) - \min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} f_a(\mathbf{x}_t) \\
&\leq \max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} \hat{f}_{m_0,a}(\mathbf{x}_t) + \frac{\epsilon_{m_0}}{2} - \min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} \hat{f}_{m_0,a}(\mathbf{x}_t) + \frac{\epsilon_{m_0}}{2} \\
&= \max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} \hat{f}_{m_0,a}(\mathbf{x}_t) - \min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)} \hat{f}_{m_0,a}(\mathbf{x}_t) + \epsilon_{m_0} \leq 2\epsilon_{m_0}.
\end{aligned}$$

Therefore by induction, this finishes the proof.

Proof of Lemma 2 We first note that for any $a \in \mathcal{A}$,

$$\begin{aligned}\mathbb{E}(N_{m,a}|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}) &\geq \frac{1}{A}\mathbb{E}\left(\sum_{t_{m-1}+1\leq t\leq t_m}\mathbb{I}\{\mathbf{x}_t\in\mathcal{R}_{m,a}\}\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\\ &= \frac{1}{A}n_m\mathbb{P}\left(\mathbf{x}_t\in\mathcal{R}_{m,a}\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\\ &\geq \frac{1}{A}n_m\mathbb{P}\left(\mathbf{x}_t\in\mathcal{Q}_{a}\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\geq\frac{p_0}{A}n_m,\end{aligned}$$

where the inequality follows from Lemma 1 and Proposition 1. By Hoeffding's inequality,

$$\begin{aligned}\mathbb{P}\left(N_{m,a}<T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)&\leq\mathbb{P}\left(\mathbb{E}(N_{m,a}|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1})-N_{m,a}>\frac{p_0}{A}n_m-T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\\ &\leq\exp\left(-\frac{2}{n_m}\left[\frac{p_0}{A}n_m-T_m\right]^2\right)\leq\exp\left(\frac{4p_0}{A}T_m-\frac{2p_0^2}{A^2}n_m\right).\end{aligned}$$

Considering the definition of n_m and T_m, when $m\leq M-1$ we have $n_m\geq\frac{2A}{p_0}T_m+\frac{A^2}{2p_0^2}\log T$, Thus,

$$\mathbb{P}\left(N_{m,a}<T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\leq\frac{1}{T}.$$

Accordingly,

$$\mathbb{P}\left(\min_{a\in\mathcal{A}}N_{m,a}<T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\leq\frac{A}{T}.$$

Proof of Lemma 3 **Step I: Characterize the estimation error for a fixed point on the grid.** We first fix $\mathbf{x}_0\in\mathcal{R}_{m,a}\cap D^C_{m,a}$ for some $a\in\mathcal{A}$. We use local polynomial regression of order β based on samples $S_{m,a}$ to estimate the conditional expected reward $f_a(\mathbf{x}_0)$:

$$\hat{f}_{m,a}(\mathbf{x}_0)=e_1^T\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)U\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)U^T\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)\right)^{-1}$$

$$\times\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)U\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)y_t\right).$$

where $U(u)=(u^r)_{|r|\leq\beta}$ is a vector-valued function from $\mathbb{R}^d$ to $\mathbb{R}^M$, $h_{m,a}=n_{m,a}^{-\frac{1}{2\beta+d}}$ is the bandwidth, e_1 is a $M_\beta\times 1$ vector whose all elements are 0 except the first one. Recall that $M_\beta=|\{\mathbf{r}:|\mathbf{r}|\leq\lfloor\beta\rfloor\}|$.

Following the construction in Hu et al. (2022), we can derive the upper bound of the estimation error for $\hat{f}_{m,a}(\mathbf{x}_0)$ as:

$$|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \leq \frac{\sqrt{M_\beta}}{\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))}(\Gamma_1 + \Gamma_2).$$

where Γ_1 and Γ_2 are defined as follows:

$$\Gamma_1 = \frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(y_t - f_a(\mathbf{x}_t))K\Big(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\Big),$$

$$\Gamma_2 = \frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(f_a(\mathbf{x}_t) - f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0))K\Big(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\Big).$$

Step IV: error bound for a fixed point

According to Lemma EC.6, with high probability $1 - 2M_\beta^2\exp\big\{-4C_{\mathcal{A}}(1 + L\sqrt{d})^2 n_{m,a}^{\frac{2\beta}{d+2\beta}}\big\}$, $\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0)) \geq \lambda_0$. Applying Bernstein's inequality specifically to the batch-dependent samples $S_{m,a}$ and following similar strategy in Hu et al. (2022), for any $\epsilon > 0$, we bound $|\Gamma_1|$ and $|\Gamma_2|$ separately as

$$\mathbb{P}(|\Gamma_1| \geq \epsilon \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \leq 2\exp\big(-\frac{3p_0 n_{m,a}h_{m,a}^d\epsilon^2}{6Av_d p_{\max} + 2p_0\epsilon}\big),$$

$$\begin{aligned}\mathbb{P}(|\Gamma_2| \geq \epsilon + \frac{A}{p_0}Lv_d p_{\max}h_{m,a}^\beta \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \leq \\ 2\exp\big(-\frac{3p_0 n_{m,a}h_{m,a}^d\epsilon^2}{6AL^2 v_d p_{\max}h_{m,a}^{2\beta} + 2L(Av_d p_{\max} + p_0)h_{m,a}^\beta\epsilon}\big).\end{aligned}$$

Apply the results above with $\epsilon = \frac{\lambda_0}{3\sqrt{M_\beta}}\epsilon_m$, we can get that

$$\begin{aligned}&\mathbb{P}\left(|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \geq \frac{\sqrt{M_\beta}}{\lambda_0}\left(\frac{2\lambda_0}{3\sqrt{M_\beta}}\epsilon_m + \frac{A}{p_0}Lv_d p_{\max}h_{m,a}^\beta\right)|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \\ &\leq 2M_\beta^2\exp\left(-\frac{3p_0 n_{m,a}h_{m,a}^d\lambda_0^2}{6AM_\beta^4 p_{\max}v_d + 2p_0\lambda_0 M_\beta^2}\right) + 2\exp\left(-\frac{p_0 n_{m,a}h_{m,a}^d\lambda_0^2\epsilon_m^2}{18AMv_d p_{\max} + 2\sqrt{M_\beta}p_0\epsilon_m\lambda_0}\right) \\ &+ 2\exp\left(-\frac{p_0 n_{m,a}h_{m,a}^d\lambda_0^2\epsilon_m^2}{18AM_\beta L^2 v_d p_{\max}h_{m,a}^{2\beta} + 2\sqrt{M_\beta}L(Av_d p_{\max} + p_0)h_{m,a}^\beta\epsilon_m\lambda_0}\right).\end{aligned}$$

Note that $\frac{A}{p_0}Lv_d p_{\max}h_{m,a}^\beta \leq \frac{\lambda_0}{3\sqrt{M_\beta}}\epsilon_m$ under the assumption that $T \geq C_1 \vee C_2 \vee \exp(\frac{18AM_\beta L^2 v_d^2 p_{\max}^2 C_{\mathcal{A}}(2\beta+d)}{p_0^2\lambda_0^2(2\beta+d+\beta d)})$ and event $\mathcal{M}_m$ holds, we can derive that $n_{m,a} \geq T_m \geq \left(\frac{3\sqrt{2AM_\beta Lv_d p_{\max}}}{p_0\lambda_0\epsilon_m}\right)^{\frac{2\beta+d}{\beta}}$ for any $a \in \mathcal{A}$. Thus,

$$\frac{\sqrt{M_\beta}}{\lambda_0}\left(\frac{2\lambda_0}{3\sqrt{M_\beta}}\epsilon_m + \frac{A}{p_0}Lv_d p_{\max}h_{m,a}^\beta\right) \leq \epsilon_m.$$

After denoting

$$C_{\mathcal{A}} = \frac{1}{4(1+L\sqrt{d})^2} \min\left\{ \frac{3p_0\lambda_0^2}{6AM_\beta^4 p_{\max} v_d + 2p_0\lambda_0 M_\beta^2},\ \frac{p_0\lambda_0^2}{18AM_\beta v_d p_{\max} + 2\sqrt{M_\beta}p_0\lambda_0}, \right.$$

$$\left. \frac{p_0\lambda_0^2}{18AM_\beta L^2 v_d p_{\max} + 2\sqrt{M_\beta}L(Av_d p_{\max} + p_0)\lambda_0} \right\}.$$

it is easy to verify that

$$\mathbb{P}\left(|\hat{f}_{m,a}(x_0) - f_a(x_0)| \geq \frac{\epsilon_m}{2} | \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \leq (2M_\beta^2 + 4)\exp\left(-C_{\mathcal{A}} n_{m,a} h_{m,a}^d \epsilon_m^2\right)$$

$$= (4 + 2M_\beta^2)\exp\left(-C_{\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2\right).$$

Moreover, marginalizing over the history $\mathcal{F}_{m-1}$ gives

$$\mathbb{P}\left(|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \geq \frac{\epsilon_m}{2} | \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}\right) \leq (4 + 2M_\beta^2)\exp\left(-C_{\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2\right).$$

Step V: uniform convergence

To extend the bound from a fixed $\mathbf{x}_0$ to the entire region $\mathcal{R}_{m,a} \cap D_{m,a}^C$, we use a standard union bound over the grid G with $|G \cap \mathcal{R}_{m,a} \cap D_{m,a}^C| \leq |G| = \delta^{-d}$. Specifically, for any $\mathbf{x}$, $|\hat{f}_{m,a}(\mathbf{x}) - f_a(\mathbf{x})| \leq |\hat{f}_{m,a}(g(\mathbf{x})) - f_a(g(\mathbf{x}))| + \frac{1}{2}L\sqrt{d}\delta$.

Therefore,

$$\begin{aligned}
&\mathbb{P}\Big(\sup_{a\in\mathcal{A}} \sup_{\mathbf{x}\in\mathcal{R}_{m,a}\cap D_{m,a}^C} \big|\hat{f}_{m,a}(\mathbf{x}) - f_{m,a}(\mathbf{x})\big| \leq \frac{\epsilon_m}{2}(1+L\sqrt{d}) \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_m, N_{m,a} = n_{m,a}\Big) \\
&\leq \mathbb{P}\Big(\sup_{a\in\mathcal{A}} \sup_{\mathbf{x}\in G\cap\mathcal{R}_{m,a}\cap D_{m,a}^C} \big|\hat{f}_{m,a}(\mathbf{x}) - f_{m,a}(\mathbf{x})\big| \leq \frac{\epsilon_m}{2} \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_m, N_{m,a} = n_{m,a}\Big) \\
&\leq A\big(4 + 2M_\beta^2\big)\exp\Big(-C_{\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2\Big).
\end{aligned}$$

Therefore, under event $\mathcal{M}_m$ we have $N_{m,a} \geq T_m$, and condisder the definition of $\mathcal{G}_m$ we have:

$$\begin{aligned}
\mathbb{P}(\mathcal{G}_m^C \mid \bar{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_m) &\leq A\delta^{-d}(4 + 2M_\beta^2)\exp(-C_{\mathcal{A}} \min_{a\in\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2) \\
&\leq A\delta^{-d}(4 + 2M_\beta^2)\exp(-C_{\mathcal{A}} T_m^{\frac{2\beta}{2\beta+d}} \epsilon_m^2) = \frac{A(4 + 2M_\beta^2)}{T}.
\end{aligned}$$

2. Proofs of Auxiliary Lemmas

Proof of Lemma EC.1. This is obvious according to the definition of weak and strong (c_0, r_0)-regularity.

Proof of Lemma EC.2. Considering our construction of the batches,

$$n_M = T - \sum_{m=1}^{M-1} n_m \geq T - \frac{A^2}{2p_0^2}(M-1)\log T - \sum_{m=1}^{M-1} \frac{1}{2M} T^{\frac{1-\gamma^m}{1-\gamma^M}} \geq T - \frac{M-1}{2M}T - \frac{A^2}{2p_0^2} M\log T + \frac{A^2}{2p_0^2}\log T$$

$$\geq \frac{A^2}{2p_0^2}\log T + \frac{T}{2} - \frac{A^2}{2p_0^2} M \log T.$$

For $M = O(\log T)$, there exists some positive constant C such that when $T \geq C$, we have $\frac{T}{4} \geq \frac{A^2}{2p_0^2} M \log T$. Thus, $n_M \geq \frac{T}{4} + \frac{A^2}{2p_0^2}\log T$, when $T \geq C$.

Proof of Lemma EC.3. For any $a \in \mathcal{A}$ and $m \in [M]$, we have $H_{m,a} \geq n_{m,a}^{-\frac{1}{d+2\beta}} \geq T^{-\frac{1}{d+2\beta}}$ and $\delta = T^{-\frac{\beta}{d+2\beta}}(\log T)^{-1}$. Therefore, $H_{m,a}\delta^{-1} \geq T^{\frac{\beta-1}{d+2\beta}} \log T$. By the smoothness condition $\beta \geq 1$, there exists C_1 such that for $T \geq C_1$, we have $H_{m,a} \geq \sqrt{d}\delta$, for convenience, we denote C_1 as $C \vee C_1$. Given $\overline{\mathcal{M}}_m$, we have

$$H_{m,a} = N_{m,a}^{-\frac{1}{d+2\beta}} \leq T_m^{-\frac{1}{d+2\beta}} \leq T_1^{-\frac{1}{d+2\beta}}.$$

As $T_m = \frac{p_0}{4AM} T^{\frac{1-\gamma^m}{1-\gamma^M}}$ and $M = O(\log T)$, we have $T_1^{-\frac{1}{d+2\beta}} \to 0$ when $T \to \infty$. That is, there must exist some positive constant C_2 such that when $T \geq C_2$, $H_{m,a} = N_{m,a}^{-\frac{1}{d+2\beta}} \leq 2r_0$.

Proof of Lemma EC.4. We proof this lemma by contradiction. Suppose there exists a point $\mathbf{x} \in D_{m,a}$ such that $\mathcal{R}_{m,a}$ is strongly (c_0, r_0)-reugular at $\mathbf{x}$.Since $\|g(\mathbf{x}) - \mathbf{x}\| \leq \frac{1}{2}\sqrt{d\delta} \leq \frac{1}{2}H_{m,a}$,

$$\mathcal{B}(\mathbf{x}, \frac{H_{m,a}}{2}) \subseteq \mathcal{B}(\mathbf{x}, H_{m,a} - \frac{1}{2}\sqrt{d\delta}) \subseteq \mathcal{B}(g(\mathbf{x}), H_{m,a}).$$

Thus,

$$\begin{aligned} Leb\left[\mathcal{B}(g(\mathbf{x}), H_{m,a}) \cap \mathcal{R}_{m,a}\right] &\geq Leb\left[\mathcal{B}(\mathbf{x}, \frac{H_{m,a}}{2}) \cap \mathcal{R}_{m,a}\right] \\ &\geq c_0 v_d 2^{-d} H_{m,a}^d = \frac{c_0}{2^d} Leb[\mathcal{B}(g(\mathbf{x}), H_{m,a})], \end{aligned}$$

where v_d is the volume of a unit ball in $\mathbb{R}^d$. Here the second inequality uses the assumption that $\mathcal{R}_{m,a}$ is strongly (c_0, r_0)-regular at $\mathbf{x}$ and that $H_{m,a}/2 \leq r_0$.

This means that if there exists any point $\mathbf{x} \in D_{m,a}$ such that $\mathcal{R}_{m,a}$ is strongly (c_0, r_0)-regular at $\mathbf{x}$, then $\mathcal{R}_{m,a}$ must be weakly $\left(\frac{c_0}{2^d}, H_{m,a}\right)$-regular at $g(\mathbf{x})$, the center of the hypercube that $\mathbf{x}$ belongs to. By construction, $\mathcal{R}_{m,a}$ is not weakly $\left(\frac{c_0}{2^d}, H_{m,a}\right)$-regular at any center of any hypercube in $D_{m,a}$. Thus this lemma holds.

Proof of Lemma EC.5. When $m = 1$, the conclusions hold trivially since $\mathcal{R}_{m,a} = \mathcal{X}$ for any $a \in \mathcal{A}$, this indicates that, we pull each arm with equal probability for all samples in the first stage. This implies that

$p_{a,1} = p(\mathbf{x})$ for $\mathbf{x} \in \mathcal{X}$. Then the conclusion follows from Assumption 3. Denote $\mathcal{T}_{m,a} = \{t_{m-1}+1 \le t \le t_m : a_t = a\}$, $N_{m,[A]} = \{N_{m,a}\}_{a\in[A]}$, and $n_{m,[A]} = \{n_{m,a}\}_{a\in[A]}$.

When $m \ge 2$, $\{(\mathbf{x}_t, y_t), t \in \mathcal{T}_{m,a}\}$ are obviously i.i.d conditionally on $\mathcal{F}_{m-1}$, $\overline{\mathcal{G}}_{m-1}$, and $\overline{\mathcal{M}}_{m-1}$, since a_t only depends on $\mathbf{x}_t$ and $\mathcal{F}_{m-1}$. This implies that $\{\mathbf{x}_t : t \in \mathcal{T}_{m,a}\}$ are i.i.d conditionally on $\mathcal{F}_{m-1}$, $\overline{\mathcal{G}}_{m-1}$, $\overline{\mathcal{M}}_{m-1}$. Furthermore, for any $\mathbf{x} \in \mathcal{X}$, it is obvious that $p(\mathbf{x}) = p_{\mathbf{x}_t|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},\mathcal{F}_{m-1}}(\mathbf{x})$ thus,

$$\begin{aligned} p_{m,a}(\mathbf{x}) &= p_{\mathbf{x}_t|a_t=a,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},\mathcal{F}_{m-1}}(\mathbf{x}) \\ &= \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p_{\mathbf{x}_t|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},\mathcal{F}_{m-1}}(\mathbf{x}) \\ &= \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p(\mathbf{x}). \end{aligned}$$

For any $\mathbf{x} \notin \mathcal{R}_{m,a}$, our algorithm ensures that $\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) = 0$, which also means $p_{m,a}(\mathbf{x}) = 0$. For any $\mathbf{x} \in \mathcal{R}_{m,a}$, our algorithm ensures that

$$\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \ge \frac{1}{A}.$$

Therefore, we can directly derive that

$$p_{m,a}(x) = \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p(\mathbf{x}) \ge \frac{p(\mathbf{x})}{A\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} \ge \frac{p_{\min}}{A},$$

where the last inequality follows from Assumption 3 and $\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \le 1$. Plus, since $T \ge C_1 \vee C_2$, Lemma 1 implies that $\mathcal{Q}_a \subseteq \mathcal{R}_{m,a}$. Thus it follows from Assumptions 2 and 3 that

$$\begin{aligned} \mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) &= \mathbb{P}(a_t = a|\mathbf{x}_t \in \mathcal{R}_{m,a}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \\ &\quad \times \mathbb{P}(\mathbf{x}_t \in \mathcal{R}_{m,a}|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \\ &\quad \ge \frac{1}{A}\mathbb{P}(\mathbf{x}_t \in \mathcal{Q}_a) \ge \frac{p_0}{A}, \end{aligned}$$

where the last inequality follows from Proposition 1. Then we can derive that

$$p_{m,a}(\mathbf{x}) = \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p(\mathbf{x}) \le \frac{Ap(\mathbf{x})}{p_0} \le \frac{Ap_{\max}}{p_0},$$

where the last inequality follows from Assumption 3, this finishes the proof.

Proof of Lemma EC.6. In the following proof, we condition on $\overline{\mathcal{G}}_{m-1}$, $\overline{\mathcal{M}}_{m-1}$, $N_{m,a} = n_{m,a}$, and $\mathcal{F}_{m-1}$. According to Lemma 2, the samples $S_{m,a} = \{(\mathbf{x}_t, y_t) : a_t = a, t_{m-1}+1 \le t \le t_m\} = \{(\mathbf{x}_t, y_t) : t \in \mathcal{T}_{m,a}\}$ are i.i.d whose conditional density for $\mathbf{x}_t$ is $p_{m,a} : \frac{1}{A} p_{\min} \le p_{m,a}(\mathbf{x}) \le \frac{Ap_{\max}}{p_0}$ for any $\mathbf{x} \in \mathcal{R}_{m,a}$.

Moreover, recall that $f_a(\mathbf{x}) = \mathbb{E}[y_t|\mathbf{x}_t = \mathbf{x}, a_t = a]$ for any $\mathbf{x} \in \mathcal{X}$ and $a \in \mathcal{A}$. Here the purpose of conditioning on $\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}$ is merely to guarantee the strong density condition for $p_{m,a}$. We fix $\mathbf{x}_0 \in G \cap \mathcal{R}_{m,a} \cap D_{m,a}^C$.

Recall that

$$\hat{\Sigma}_{m,a}(\mathbf{x}_0; S_{m,a}) = \frac{1}{n_{m,a} h_{m,a}^d} \sum_{t \in \mathcal{T}_{m,a}} K\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right) U\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right) U^T\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right).$$

where $U(u) = (u^r)_{|r| \le \beta}$ is a vector-valued function from $\mathbb{R}^d$ to $\mathbb{R}^M$, $h_{m,a} = n_{m,a}^{-\frac{1}{2\beta+d}}$ is the bandwidth, e_1 is a $M \times 1$ vector whose all elements are 0 except the first one.

Note that the (r_1, r_2)-th entry of $\hat{\Sigma}_{m,a}(\mathbf{x}_0)$ is

$$(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} = \frac{1}{n_{m,a} h_{m,a}^d} \sum_{t \in \mathcal{T}_{m,a}} \left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right)^{r_1+r_2} K\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right)$$

whose conditional expectation is

$$\begin{aligned}
&\mathbb{E}[(\hat{\Sigma}_{m,a}(\mathbf{x}_0)_{r_1,r_2}|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}] \\
&= \frac{1}{h_{m,a}^d} \int_{\|\frac{\mathbf{x}-\mathbf{x}_0}{h_{m,a}}\| \le 1, \mathbf{x} \in \mathcal{R}_{m,a}} \left(\frac{\mathbf{x} - \mathbf{x}_0}{h_{m,a}}\right)^{r_1+r_2} p_{m,a}(\mathbf{x}) dx \\
&= \int_{\|u\| \le 1, \mathbf{x}_0 + u h_{m,a} \in \mathcal{R}_{m,a}} u^{s_1+s_2} p_{m,a}(\mathbf{x}_0 + h_{m,a}\mathbf{u}) du \\
&:= (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}
\end{aligned}$$

It follows that

$$\begin{aligned}
\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0) &\ge \min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W + \min_{\|W\|=1} W^T(\hat{\Sigma}_{m,a}(\mathbf{x}_0) - \Sigma_{m,a}(\mathbf{x}_0)) W \\
&\ge \min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W - \sum_{|r_1|, |r_2| \le \lfloor \beta \rfloor} |(\hat{\Sigma}_{m,a}(\mathbf{x}_0)_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}|.
\end{aligned}$$

We first derive lower bound for $\min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W$:

$$\begin{aligned}
W^T \Sigma_{m,a}(\mathbf{x}_0) W &= \int_{B_{m,a}(\mathbf{x}_0)} \left(\sum_{|r| \le \lfloor \beta \rfloor} W_r u^r\right)^2 p_{m,a}(\mathbf{x}_0 + h_{m,a} u) du \\
&\ge \frac{1}{A} p_{\min} \int_{B_{m,a}(\mathbf{x}_0)} \left(\sum_{|r| \le \lfloor \beta \rfloor} W_r u^r\right)^2 du,
\end{aligned}$$

where $B_{m,a}(\mathbf{x}_0) = \{u \in \mathbb{R}^d : \|u\| < 1, \mathbf{x}_0 + uh_{m,a} \in \mathcal{R}_{m,a}\}$.

Note that

$$\begin{aligned}
\mathrm{Leb}[B_{m,a}(\mathbf{x}_0)] &= h_{m,a}^{-d}\mathrm{Leb}[\mathcal{B}(\mathbf{x}_0, h_{m,a}) \cap \mathcal{R}_{m,a}] \\
&\geq h_{m,a}^{-d} c_0 \mathrm{Leb}[\mathcal{B}(\mathbf{x}_0, h_{m,a})] = h_{m,a}^{-d} c_0 v_d h_{m,a}^d = c_0 v_d
\end{aligned}$$

where the first equality holds because of change of variable, and the inequality holds since $\mathbf{x}_0 \notin D_{m,a}$ and our construction of $D_{m,a}$ guarantees that $\mathcal{R}_{m,a}$ is weakly $(c_0, h_{m,a})$-regular at $\mathbf{x}_0 \in \mathcal{R}_{m,a} \cap D_{m,a}^C$. Furthermore, as $\mathcal{R}_{m,a}$ is compact, we have

$$\begin{aligned}
\lambda_{\min}(\Sigma_{m,a}(\mathbf{x}_0)) &= \min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W \\
&\geq \frac{1}{A} p_{\min} \min_{\|W\|=1: S \subseteq \mathcal{B}(0,1), \mathrm{Leb}(S) = c_0} \int_S \left(\sum_{|s| \leq \lfloor \beta \rfloor} W_s u^s \right)^2 du = 2\lambda_0 > 0.
\end{aligned}$$

Now we derive upper bound for $\sum_{|r_1|,|r_2| \leq \lfloor \beta \rfloor} |(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}|$.

For $t \in S_{m,a}$ and $|r_1|, |r_2| \leq \beta$, denote $Z_t(r_1, r_2) = \frac{1}{h_{m,a}^d} \left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}} \right)^{r_1 + r_2} K\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}} \right)$. Obviously, $|Z_t(r_1, r_2)| \leq \frac{1}{h_{m,a}^d}$, and

$$\begin{aligned}
&\mathbb{E}(Z_t^2(r_1, r_2) | a_t = a, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \\
&= \frac{1}{h_{m,a}^{2d}} \int_{\mathcal{X}} \left(\frac{\mathbf{x} - \mathbf{x}_0}{h_{m,a}} \right)^{2(r_1 + r_2)} K^2\left(\frac{\mathbf{x} - \mathbf{x}_0}{h_{m,a}} \right) p_{m,a}(\mathbf{x}) dx \\
&\leq \frac{A p_{\max}}{p_0 h_{m,a}^d} \int_{B_{m,a}(\mathbf{x}_0)} u^{2(r_1 + r_2)} du \leq \frac{A p_{\max}}{p_0 h_{m,a}^d} \mathrm{Leb}[B_{m,a}(\mathbf{x}_0)] \leq \frac{A p_{\max} v_d}{p_0 h_{m,a}^d},
\end{aligned}$$

where the first inequality follows from the fact that $p_{m,a}(\mathbf{x}) \leq \frac{A p_{\max}}{p_0}$.

By Bernstein inequality,

$$\begin{aligned}
&\mathbb{P}\left(|(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}| \geq \frac{\lambda_0}{M_\beta^2} \,\middle|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \\
&= \mathbb{P}\left(\left|\sum_{t\in\mathcal{T}_{m,a}} Z_t(r_1,r_2) - n_{m,a}\mathbb{E}\left(\sum_{t\in\mathcal{T}_{m,a}} Z_t(r_1,r_2) \,\middle|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right)\right|\right. \\
&\quad \left. \geq n_{m,a}\frac{\lambda_0}{M_\beta^2} \,\middle|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \\
&\leq 2\exp\left(-\frac{3p_0 n_{m,a} h_{m,a}^d \lambda_0^2}{6AM_\beta^4 p_{\max} v_d + 2p_0\lambda_0 M_\beta^2}\right).
\end{aligned}$$

By taking union bound over all possible r_1, r_2,

$$\begin{aligned}
&\mathbb{P}\left(\sum_{|r_1|,|r_2|\leq\lfloor\beta\rfloor} |(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}| \geq \lambda_0 \,\middle|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \\
&\leq 2M_\beta^2 \exp\left(-\frac{3p_0 n_{m,a} h_{m,a}^d \lambda_0^2}{6AM_\beta^4 p_{\max} v_d + 2p_0\lambda_0 M_\beta^2}\right) \\
&\leq 2M_\beta^2 \exp\left\{-4C_{\mathcal{A}}(1+L\sqrt{d})^2 n_{m,a} h_{m,a}^d\right\} \\
&\leq 2M_\beta^2 \exp\left\{-4C_{\mathcal{A}}(1+L\sqrt{d})^2 n_{m,a}^{\frac{2\beta}{2\beta+d}}\right\}
\end{aligned}$$

According to these results, with high probability $1 - 2M_\beta^2 \exp\left\{-4C_{\mathcal{A}}(1+L\sqrt{d})^2 n_{m,a}^{\frac{2\beta}{2\beta+d}}\right\}$, we have $\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0)) \geq \lambda_0$.

Proof of Lemma EC.7. When $m = 1$, the conclusions hold trivially since $\mathcal{R}_{m,a} = \mathcal{X}$ for any $a \in \mathcal{A}$, i.e., we pull each arm with equal probability for all samples in the first stage. This implies that $p_{a,1} = p(\mathbf{x})$ for $\mathbf{x} \in \mathcal{X}$. Then the conclusion follows from Assumption 3. Denote $\mathcal{T}_{m,a} = \{t_{m-1}+1 \leq t \leq t_m : a_t = a\}$, $N_{m,[A]} = \{N_{m,a}\}_{a\in[A]}$, and $n_{m,[A]} = \{n_{m,a}\}_{a\in[A]}$.

When $m \geq 2$, $\{(\mathbf{x}_t, y_t), t \in \mathcal{T}_{m,a}\}$ are obviously i.i.d conditionally on $\mathcal{F}_{m-1}$, $\overline{\mathcal{G}}_{m-1}$, and $\overline{\mathcal{M}}_{m-1}$, since a_t only depends on $\mathbf{x}_t$ and $\mathcal{F}_{m-1}$. This implies that $\{\mathbf{x}_t : t \in \mathcal{T}_{m,a}\}$ are i.i.d conditionally on $\mathcal{F}_{m-1}$, $\overline{\mathcal{G}}_{m-1}$, $\overline{\mathcal{M}}_{m-1}$. Furthermore, for any $\mathbf{x} \in \mathcal{X}$, it is obvious that $p(\mathbf{x}) = p_{\mathbf{x}_t|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},\mathcal{F}_{m-1}}(\mathbf{x})$ thus,

$$\begin{aligned}
p_{m,a}(\mathbf{x}) &= p_{\mathbf{x}_t|a_t=a,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},\mathcal{F}_{m-1}}(\mathbf{x}) \\
&= \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p_{\mathbf{x}_t|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},\mathcal{F}_{m-1}}(\mathbf{x}) \\
&= \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p(\mathbf{x}).
\end{aligned}$$

For any $\mathbf{x} \notin \mathcal{R}_{m,a}$, our algorithm ensures that $\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) = 0$, which also means $p_{m,a}(\mathbf{x}) = 0$. For any $\mathbf{x} \in \mathcal{R}_{m,a}$, our algorithm ensures that

$$\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \geq \frac{1}{A}.$$

Therefore, we can directly derive that

$$p_{m,a}(\mathbf{x}) = \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p(\mathbf{x}) \geq \frac{p(\mathbf{x})}{A\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} \geq \frac{p_{\min}}{A},$$

where the last inequality follows from Assumption 3 and $\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \leq 1$. Plus, Lemma EC.9 implies that $\mathcal{Q}_a \subseteq \mathcal{R}_{m,a}$. Thus it follows from Assumption 3 that

$$\begin{aligned}
\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) &= \mathbb{P}(a_t = a|\mathbf{x}_t \in \mathcal{R}_{m,a}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \\
&\quad \times \mathbb{P}(\mathbf{x}_t \in \mathcal{R}_{m,a}|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}) \\
&\geq \frac{1}{A}\mathbb{P}(\mathbf{x}_t \in \mathcal{Q}_a) \geq \frac{1}{A}\mathbb{P}(\mathbf{x}_t \in \mathcal{Q}_a \cap B(\mathbf{x}_0, r_0)) \geq \frac{p_0}{A},
\end{aligned}$$

where the last inequality follows from Assumption 6. Then we can derive that

$$p_{m,a}(\mathbf{x}) = \frac{\mathbb{P}(a_t = a|\mathbf{x}_t = \mathbf{x}, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})}{\mathbb{P}(a_t = a|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1})} p(\mathbf{x}) \leq \frac{Ap(\mathbf{x})}{p_0} \leq \frac{Ap_{\max}}{p_0},$$

where the last inequality follows from Assumption 3, this finishes the proof.

Proof of Lemma EC.8. In the following proof, we condition on $\overline{\mathcal{G}}_{m-1}$, $\overline{\mathcal{M}}_{m-1}$, $N_{m,a} = n_{m,a}$, and $\mathcal{F}_{m-1}$. Recall that $f_a(\mathbf{x}) = \mathbb{E}[y_t|\mathbf{x}_t = \mathbf{x}, a_t = a]$ for any $\mathbf{x} \in \mathcal{X}$ and $a \in \mathcal{A}$. Here the purpose of conditioning on $\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, \mathcal{F}_{m-1}$ is merely to guarantee the strong density condition for $p_{m,a}$. We fix $\mathbf{x}_0 \in \mathcal{R}_{m,a}$.

Recall that

$$\hat{\Sigma}_{m,a}(\mathbf{x}_0; S_{m,a}) = \frac{1}{n_{m,a}h_{m,a}^d} \sum_{t \in \mathcal{T}_{m,a}} K\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right) U\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right) U^T\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right).$$

where $U(u) = (u^r)_{|r| \leq \beta}$ is a vector-valued function from $\mathbb{R}^d$ to $\mathbb{R}^M$, $h_{m,a} = n_{m,a}^{-\frac{1}{2\beta+d}}$ is the bandwidth, e_1 is a $M_\beta \times 1$ vector whose all elements are 0 except the first one.

Note that the (r_1, r_2)-th entry of $\hat{\Sigma}_{m,a}(\mathbf{x}_0)$ is

$$(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} = \frac{1}{n_{m,a}h_{m,a}^d} \sum_{t \in \mathcal{T}_{m,a}} \left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right)^{r_1+r_2} K\left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}}\right)$$

whose conditional expectation is

$$\begin{aligned}
(\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2} &:= \mathbb{E}[(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2}|\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}] \\
&= \mathbb{E}[(\mathbf{x}_t - \mathbf{x}_0)^{r_1+r_2}|\mathbf{x}_t \in \mathcal{R}_{m,a} \cap \mathcal{B}(\mathbf{x}_0, h_{m,a})]
\end{aligned}$$

It follows that

$$\begin{aligned}\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0)) &\geq \min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W + \min_{\|W\|=1} W^T (\hat{\Sigma}_{m,a}(\mathbf{x}_0) - \Sigma_{m,a}(\mathbf{x}_0)) W \\ &\geq \min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W - \sum_{|r_1|,|r_2| \leq \lfloor \beta \rfloor} |(\hat{\Sigma}_{m,a}(\mathbf{x}_0)_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}|.\end{aligned}$$

We first derive lower bound for $\min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W$. Similar to Lemma EC.3, we can derive that there exist some positive C_3 such that $h_{m,a} \leq r_0$ when $T \geq C_3$. Thus $\mathcal{B}(\mathbf{x}_0, h_{m,a}) \subseteq \mathcal{B}(\mathbf{x}_0, r_0)$. As $\mathbf{x}_t \in \mathcal{R}_{m,a} \cap \mathcal{B}(\mathbf{x}_0, r_0)$, $Q_a \subseteq \mathcal{R}_{m,a}$ and according to Assumption 6, we have

$$\min_{\|W\|=1} W^T \Sigma_{m,a}(\mathbf{x}_0) W \geq \lambda_{min}(\Sigma_{m,a}(\mathbf{x}_0)) \geq \widetilde{\lambda}_0.$$

Now we derive upper bound for $\sum_{|r_1|,|r_2| \leq \lfloor \beta \rfloor} |(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}|$. For $t \in S_{m,a}$ and $|r_1|, |r_2| \leq \beta$, denote $Z_t(r_1, r_2) = \frac{1}{h_{m,a}^d} \left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}} \right)^{r_1 + r_2} K \left(\frac{\mathbf{x}_t - \mathbf{x}_0}{h_{m,a}} \right)$. Obviously, $|Z_t(r_1, r_2)| \leq \frac{1}{h_{m,a}^d}$, and

$$\begin{aligned}&\mathbb{E}(Z_t^2(r_1, r_2) | a_t = a, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \\ &= \frac{1}{h_{m,a}^{2d}} \int_{\mathcal{X}} \left(\frac{\mathbf{x} - \mathbf{x}_0}{h_{m,a}} \right)^{2(r_1 + r_2)} K^2 \left(\frac{\mathbf{x} - \mathbf{x}_0}{h_{m,a}} \right) p_{m,a}(\mathbf{x}) dx \\ &\leq \frac{A p_{\max}}{p_0 h_{m,a}^d} \int_{B_{m,a}(\mathbf{x}_0)} u^{2(r_1 + r_2)} du \leq \frac{A p_{\max}}{p_0 h_{m,a}^d} \mathrm{Leb}[B_{m,a}(\mathbf{x}_0)] \leq \frac{A p_{\max} v_d}{p_0 h_{m,a}^d},\end{aligned}$$

where the first inequality follows from the fact that $p_{m,a}(\mathbf{x}) \leq \frac{A p_{\max}}{p_0}$.

By Bernstein inequality,

$$\begin{aligned}&\mathbb{P}\left(|(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2} - (\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}| \geq \frac{\lambda_0}{M_\beta^2} \middle| \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1} \right) \\ &= \mathbb{P}\left(\left| \sum_{t \in \mathcal{T}_{m,a}} Z_t(r_1, r_2) - n_{m,a} \mathbb{E}\left(\sum_{t \in \mathcal{T}_{m,a}} Z_t(r_1, r_2) \middle| \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1} \right) \right| \right. \\ &\qquad \left. \geq n_{m,a} \frac{\lambda_0}{M_\beta^2} \middle| \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1} \right) \\ &\leq 2 \exp\left(-\frac{3 p_0 n_{m,a} h_{m,a}^d \lambda_0^2}{6 A M_\beta^4 p_{\max} v_d + 2 p_0 \lambda_0 M_\beta^2} \right).\end{aligned}$$

By taking union bound over all possible r_1, r_2,

$$\begin{aligned}
&\mathbb{P}\left(\sum_{|r_1|,|r_2|\le\lfloor\beta\rfloor}|(\hat{\Sigma}_{m,a}(\mathbf{x}_0))_{r_1,r_2}-(\Sigma_{m,a}(\mathbf{x}_0))_{r_1,r_2}|\ge\widetilde{\lambda}_0\,\middle|\,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},N_{m,a}=n_{m,a},\mathcal{F}_{m-1}\right)\\
&\le 2M_\beta^2\exp\left(-\frac{3p_0 n_{m,a}h_{m,a}^d\widetilde{\lambda}_0^2}{6AM_\beta^4 p_{\max}v_d+2p_0\widetilde{\lambda}_0M_\beta^2}\right)\\
&\le 2M_\beta^2\exp\left\{-4\widetilde{C}_{\mathcal{A}}n_{m,a}h_{m,a}^d\right\}\\
&\le 2M_\beta^2\exp\left\{-4\widetilde{C}_{\mathcal{A}}n_{m,a}^{\frac{2\beta}{2\beta+d}}\right\}
\end{aligned}$$

According to these results, with high probability $1-2M_\beta^2\exp\left\{-4\widetilde{C}_{\mathcal{A}}n_{m,a}^{\frac{2\beta}{2\beta+d}}\right\}$, we have $\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))\ge\widetilde{\lambda}_0$.

Proof of Lemma EC.9. We prove this lemma by induction. When $m=1$, it is trivially true that $Q_a\subseteq\mathcal{R}_{1,a}=\mathcal{X}$ and $\max_{a\in\mathcal{A}}f_a(\mathbf{x}_t)-\min_{a\in\mathcal{A}}f_a(\mathbf{x}_t)\le 2$. Assume that the lemma holds for $m\le m_0$, we need to proof that it also holds for m_0+1. Under the event $\overline{\mathcal{G}}_{m_0}\cap\overline{\mathcal{M}}_{m_0}$, if $\mathbf{x}_t\in Q_a\subseteq R_{m_0,a}$, $f_a(\mathbf{x}_t)\ge\max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)}f_a(\mathbf{x}_t)$ and considering the event $\mathcal{G}_{m_0}$, $|\hat{f}_{m_0,a}(\mathbf{x}_t)-f_a(\mathbf{x}_t)|\le\frac{\epsilon_{m_0}}{2}$ we can derive that

$$\hat{f}_{m_0,a}(\mathbf{x}_t)+\frac{\epsilon_{m_0}}{2}\ge f_a(\mathbf{x}_t)\ge\max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)}f_a(\mathbf{x}_t)\ge\max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)}\hat{f}_{m_0,a}(\mathbf{x}_t)-\frac{\epsilon_{m_0}}{2}.$$

This is equal to $\hat{f}_{m_0,a}(\mathbf{x}_t)\ge\max_{a\in\mathcal{A}_{m_0}(\mathbf{x}_t)}\hat{f}_{m_0,a}(\mathbf{x}_t)-\epsilon_{m_0}$. according to the definition of active arm set $\mathcal{A}_m(\mathbf{x}_t)$ in Algorithm 2, we have $a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)$ and that is $\mathbf{x}_t\in R_{m_0+1,a}$. Thus we have $\mathcal{Q}_a\subseteq\mathcal{R}_{m_0+1,a}$.

We then proof that for m_0+1, $\max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}f_a(\mathbf{x}_t)-\min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}f_a(\mathbf{x}_t)\le 2\epsilon_{m_0}$. As $\mathcal{A}_{m_0+1}(\mathbf{x}_t)\subseteq\mathcal{A}_{m_0}(\mathbf{x}_t)$ and $\mathcal{Q}_a\subseteq\mathcal{R}_{m_0+1,a}$, then $a_t^*\in\mathcal{A}_{m+1}(\mathbf{x}_t)$ and under event $\mathcal{G}_{m_0}$, we have

$$\begin{aligned}
\max_{a\in\mathcal{A}}f_a(\mathbf{x}_t)-\min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}f_a(\mathbf{x}_t)&=\max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}f_a(\mathbf{x}_t)-\min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}f_a(\mathbf{x}_t)\\
&\le\max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}\hat{f}_{m_0,a}(\mathbf{x}_t)+\frac{\epsilon_{m_0}}{2}-\min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}\hat{f}_{m_0,a}(\mathbf{x}_t)+\frac{\epsilon_{m_0}}{2}\\
&=\max_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}\hat{f}_{m_0,a}(\mathbf{x}_t)-\min_{a\in\mathcal{A}_{m_0+1}(\mathbf{x}_t)}\hat{f}_{m_0,a}(\mathbf{x}_t)+\epsilon_{m_0}\le 2\epsilon_{m_0}.
\end{aligned}$$

Therefore by induction, this finishes the proof.

Proof of Lemma EC.10. We first note that for any $a \in \mathcal{A}$,

$$\begin{aligned}
\mathbb{E}(N_{m,a}|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}) &\geq \frac{1}{A}\mathbb{E}\left(\sum_{t_{m-1}+1\leq t\leq t_m}\mathbb{I}\{\mathbf{x}_t\in\mathcal{R}_{m,a}\}\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\\
&= \frac{1}{A}n_m\mathbb{P}\left(\mathbf{x}_t\in\mathcal{R}_{m,a}\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\\
&\geq \frac{1}{A}n_m\mathbb{P}\left(\mathbf{x}_t\in\mathcal{Q}_{a}\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\geq\frac{p_0}{A}n_m,
\end{aligned}$$

where the inequality follows from LemmaEC.9 and Proposition 1. By Hoeffding's inequality,

$$\begin{aligned}
\mathbb{P}\left(N_{m,a}<T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right) &\leq \mathbb{P}\left(\mathbb{E}(N_{m,a}|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1})-N_{m,a}>\frac{p_0}{A}n_m-T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\\
&\leq \exp\left(-\frac{2}{n_m}\left[\frac{p_0}{A}n_m-T_m\right]^2\right)\leq\exp\left(\frac{4p_0}{A}T_m-\frac{2p_0^2}{A^2}n_m\right).
\end{aligned}$$

Considering the definition of n_m and T_m, when $m\leq M-1$ we have $n_m\geq\frac{2A}{p_0}T_m+\frac{A^2}{2p_0^2}\log T$, Thus,

$$\mathbb{P}\left(N_{m,a}<T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\leq\frac{1}{T}.$$

Accordingly,

$$\mathbb{P}\left(\min_{a\in\mathcal{A}}N_{m,a}<T_m\big|\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1}\right)\leq\frac{A}{T}.$$

Proof of Lemma EC.11. **Step I: Characterize the estimation error for a fixed point on the grid.** We first fix $\mathbf{x}_0\in\mathcal{R}_{m,a}$ for some $a\in\mathcal{A}$. We use local polynomial regression of order β based on samples $S_{m,a}=\{(\mathbf{x}_t,y_t):a_t=a,t_{m-1}+1\leq t\leq t_m\}$ to estimate the conditional expected reward $f_a(\mathbf{x}_0)$:

$$\begin{aligned}
&\hat{f}_{m,a}(\mathbf{x}_0)=e_1^T\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)U\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)U^T\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)\right)^{-1}\\
&\times\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)U\left(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\right)y_t\right).
\end{aligned}$$

where $U(u)=(u^r)_{|r|\leq\beta}$ is a vector-valued function from $\mathbb{R}^d$ to $\mathbb{R}^M$, $h_{m,a}=n_{m,a}^{-\frac{1}{2\beta+d}}$ is the bandwidth, e_1 is a $M_\beta\times 1$ vector whose all elements are 0 except the first one. Recall that $M_\beta=|\{r:|r|\leq\lfloor\beta\rfloor\}|$.

According to Proposition 1.12 in Tsybakov (2008), the true conditional expected reward f_a can be written in the following way:

$$f_a(\mathbf{x}_0) = f_{a,\lfloor\beta\rfloor}(\mathbf{x}_0;\mathbf{x}_0) = e_1^T\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U^T\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\right)^{-1}$$

$$\times\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0)\right),$$

where

$$f_{a,\lfloor\beta\rfloor}(\mathbf{x};\mathbf{x}_0) = \sum_{|r|\le\lfloor\beta\rfloor}\frac{(\mathbf{x}-\mathbf{x}_0)^r}{r!}D^r f_a(\mathbf{x}_0).$$

After denoting

$$\hat{\Sigma}_{m,a}(\mathbf{x}_0) = \frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U^T\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big),$$

the estimation error for $\hat{f}_{m,a}(\mathbf{x}_0)$ has the following upper bound:

$$|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \le \left\|\left(\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U^T\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\right)^{-1}\right\|$$

$$\times\left\|\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)U\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)(y_t - f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0))\right\|$$

$$\le \frac{\sqrt{M_\beta}}{\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))}\left|\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(y_t - f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\right|$$

$$< \frac{\sqrt{M_\beta}}{\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))}\left|\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(y_t - f_a(\mathbf{x}_t))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\right|$$

$$+\frac{\sqrt{M_\beta}}{\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))}\left|\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(f_a(\mathbf{x}_t) - f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\right|,$$

where the second inequality follows from the fact that $U(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}})K(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}})$ is an $M_\beta\times 1$ vector whose elements are bounded by $K(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}})$.

We further denote

$$\Gamma_1 = \frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(y_t - f_a(\mathbf{x}_t))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big),$$

$$\Gamma_2 = \frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}(f_a(\mathbf{x}_t) - f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big).$$

Then

$$|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \le \frac{\sqrt{M_\beta}}{\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))}(\Gamma_1 + \Gamma_2).$$

Step II: lower bound for $\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))$.

According to Lemma EC.8, with high probability $1-2M_\beta^2\exp\big\{-4\widetilde{C}_{\mathcal{A}}n_{m,a}^{\frac{2\beta}{d+2\beta}}\big\}$,

$$\lambda_{\min}(\hat{\Sigma}_{m,a}(\mathbf{x}_0))\ge\widetilde{\lambda}_0.$$

Step III: upper bound Γ_1 and Γ_2.

We first bound

$$\Gamma_1=\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}\big(y_t-f_a(\mathbf{x}_t)\big)K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)=\frac{1}{n_{m,a}}\sum_{t\in\mathcal{T}_{m,a}}\hat{Z}_t,$$

where

$$\hat{Z}_t=\frac{1}{h_{m,a}^d}(y_t-f_a(\mathbf{x}_t))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big).$$

It is easy to prove that

$$|\hat{Z}_t|\le h_{m,a}^{-d},$$

$$\mathbb{E}(\hat{Z}_t\,|\,A_t=a,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},N_{m,a}=n_{m,a},\mathcal{F}_{m-1})=0,$$
$$\mathbb{E}(\hat{Z}_t^2\,|\,A_t=a,\overline{\mathcal{G}}_{k-1},\overline{\mathcal{M}}_{m-1},N_{m,a}=n_{m,a},\mathcal{F}_{m-1})\le\frac{Av_dp_{\max}}{p_0h_{m,a}^d},$$

where the last inequality uses the fact that $|y_t-f_a(\mathbf{x}_t)|\le1$. By Bernstein's inequality, for $\forall\epsilon>0$,

$$\mathbb{P}(|\Gamma_1|\ge\epsilon\,\big|\,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},N_{m,a}=n_{m,a},\mathcal{F}_{m-1})=\mathbb{P}(\big[\sum_{t\in\mathcal{T}_{m,a}}\hat{Z}_t\big]\ge n_{m,a}\epsilon\,\big|\,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},N_{m,a}=n_{m,a},\mathcal{F}_{m-1})$$

$$\le2\exp\big(-\frac{3p_0n_{m,a}h_{m,a}^d\epsilon^2}{6Av_dp_{\max}+2p_0\epsilon}\big).$$

We now bound

$$\Gamma_2=\frac{1}{n_{m,a}h_{m,a}^d}\sum_{t\in\mathcal{T}_{m,a}}\big(f_a(\mathbf{x}_t)-f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0)\big)K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)=\frac{1}{n_{m,a}}\sum_{t\in\mathcal{T}_{m,a}}\hat{Z}_t,$$

where

$$\hat{Z}_t=\frac{1}{h_{m,a}^d}(f_a(\mathbf{x}_t)-f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0))K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big).$$

Note that by the definition of Hölder class in Assumption 1,

$$\Big|\big(f_a(\mathbf{x}_t)-f_{a,\lfloor\beta\rfloor}(\mathbf{x}_t;\mathbf{x}_0)\big)K\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\Big|\le LK\Big(\frac{\mathbf{x}_t-\mathbf{x}_0}{h_{m,a}}\Big)\|\mathbf{x}_t-\mathbf{x}_0\|^\beta\le Lh_{m,a}^\beta.$$

It follows that

$$|\mathbb{E}(\hat{Z}_t|a_t=a,\overline{\mathcal{G}}_{m-1},\overline{\mathcal{M}}_{m-1},N_{m,a}=n_{m,a},\mathcal{F}_{m-1})|\le\frac{A}{p_0}Lv_dp_{\max}h_{m,a}^\beta$$

$$|\hat{Z}_t - \mathbb{E}(\hat{Z}_t|a_t = a, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1})| \leq \frac{A}{p_0} L v_d p_{\max} h_{m,a}^{\beta} + L h_{m,a}^{\beta-d}$$

$$\mathbb{E}(\hat{Z}_t^2|a_t = a, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \leq \frac{A}{p_0} L^2 v_d p_{\max} h_{m,a}^{2\beta-d}.$$

By Bernstein's inequality, for $\forall \epsilon > 0$,

$$\begin{aligned}
&\mathbb{P}(|\Gamma_2| \geq \epsilon + \frac{A}{p_0} L v_d p_{\max} h_{m,a}^{\beta} \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \\
&= \mathbb{P}(\big[\frac{1}{n_{m,a}} \sum_{t \in \mathcal{T}_{m,a}} \hat{Z}_t\big] \geq \epsilon + \frac{A}{p_0} L v_d p_{\max} h_{m,a}^{\beta} \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \\
&\leq \mathbb{P}(\frac{1}{n_{m,a}} \Big| \sum_{t \in \mathcal{T}_{m,a}} \hat{Z}_t - \mathbb{E}(\hat{Z}_t \,\big|\, \overline{\mathcal{G}}_{m-1}, N_{a,k} = n_{m,a}, \mathcal{F}_{m-1}) \Big| \geq \epsilon \,\big|\, \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}) \\
&\leq 2\exp\big(-\frac{3 p_0 n_{m,a} h_{m,a}^d \epsilon^2}{6 A L^2 v_d p_{\max} h_{m,a}^{2\beta} + 2L(A v_d p_{\max} + p_0) h_{m,a}^{\beta} \epsilon}\big).
\end{aligned}$$

Step IV: error bound for a fixed point

Apply the results above with $\epsilon = \frac{\widetilde{\lambda}_0}{3\sqrt{M_\beta}} \epsilon_m$, we can get that

$$\begin{aligned}
&\mathbb{P}\left(|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \geq \frac{\sqrt{M_\beta}}{\widetilde{\lambda}_0} \left(\frac{2\widetilde{\lambda}_0}{3\sqrt{M_\beta}} \epsilon_m + \frac{A}{p_0} L v_d p_{\max} h_{m,a}^{\beta}\right) | \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \\
&\leq 2 M_\beta^2 \exp\left(-\frac{3 p_0 n_{m,a} h_{m,a}^d \widetilde{\lambda}_0^2}{6 A M_\beta^4 p_{\max} v_d + 2 p_0 \widetilde{\lambda}_0 M_\beta^2}\right) + 2\exp\left(-\frac{p_0 n_{m,a} h_{m,a}^d \widetilde{\lambda}_0^2 \epsilon_m^2}{18 A M v_d p_{\max} + 2\sqrt{M_\beta} p_0 \epsilon_m \widetilde{\lambda}_0}\right) \\
&+ 2\exp\left(-\frac{p_0 n_{m,a} h_{m,a}^d \widetilde{\lambda}_0^2 \epsilon_m^2}{18 A M_\beta L^2 v_d p_{\max} h_{m,a}^{2\beta} + 2\sqrt{M_\beta} L (A v_d p_{\max} + p_0) h_{m,a}^{\beta} \epsilon_m \widetilde{\lambda}_0}\right).
\end{aligned}$$

Note that $\frac{A}{p_0} L v_d p_{\max} h_{m,a}^{\beta} \leq \frac{\widetilde{\lambda}_0}{3\sqrt{M_\beta}} \epsilon_m$ under the assumption that $T \geq C_3 \vee \exp(\frac{18 A M_\beta L^2 v_d^2 p_{\max}^2 \widetilde{C}_{\mathcal{A}}}{p_0^2 \widetilde{\lambda}_0^2})$ and event $\mathcal{M}_m$ holds, we can derive that $n_{m,a} \geq T_m \geq \left(\frac{3\sqrt{2 A M_\beta L v_d p_{\max}}}{p_0 \widetilde{\lambda}_0 \epsilon_m}\right)^{\frac{2\beta+d}{\beta}}$ for any $a \in \mathcal{A}$. Thus,

$$\frac{\sqrt{M_\beta}}{\widetilde{\lambda}_0} \left(\frac{2\widetilde{\lambda}_0}{3\sqrt{M_\beta}} \epsilon_m + \frac{A}{p_0} L v_d p_{\max} h_{m,a}^{\beta}\right) \leq \epsilon_m.$$

After denoting

$$\begin{aligned}
\widetilde{C}_{\mathcal{A}} = \frac{1}{4} \min\Bigg\{ &\frac{3 p_0 \widetilde{\lambda}_0^2}{6 A M_\beta^4 p_{\max} v_d + 2 p_0 \widetilde{\lambda}_0 M_\beta^2}, \ \frac{p_0 \widetilde{\lambda}_0^2}{18 A M_\beta v_d p_{\max} + 2\sqrt{M_\beta} p_0 \widetilde{\lambda}_0}, \\
&\frac{p_0 \widetilde{\lambda}_0^2}{18 A M_\beta L^2 v_d p_{\max} + 2\sqrt{M_\beta} L (A v_d p_{\max} + p_0) \widetilde{\lambda}_0} \Bigg\}.
\end{aligned}$$

it is easy to verify that

$$2 M_\beta^2 \exp\left\{-4\widetilde{C}_{\mathcal{A}}\ n_{m,a}^{2\beta+d}\right\} \leq 2 M_\beta^2 \exp\left(-4\widetilde{C}_{\mathcal{A}} (1 + L\sqrt{d})^2 n_{m,a} h_{m,a}^d \epsilon_m^2\right),$$

$$\exp\left(-\frac{p_0 n_{m,a} h_{m,a}^d \widetilde{\lambda}_0^2 \epsilon_m^2}{18AM_\beta v_d p_{\max} + 2\sqrt{M_\beta} p_0 \epsilon_m \widetilde{\lambda}_0}\right) \le \exp\left(-4\widetilde{C}_{\mathcal{A}}(1+L\sqrt{d})^2 n_{m,a} h_{m,a}^d \epsilon_m^2\right),$$

$$\exp\left(-\frac{p_0 n_{m,a} h_{m,a}^d \widetilde{\lambda}_0^2 \epsilon_m^2}{18AM_\beta L^2 v_d p_{\max} h_{m,a}^{2\beta} + 2\sqrt{M_\beta} L(Av_d p_{\max}+p) h_{m,a}^{\beta} \epsilon_m \widetilde{\lambda}_0}\right) \le \exp\left(-4\widetilde{C}_{\mathcal{A}}(1+L\sqrt{d})^2 n_{m,a} h_{m,a}^d \epsilon_m^2\right).$$

Therefore,

$$\mathbb{P}\left(|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \ge \frac{\epsilon_m}{2} |\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}, \mathcal{F}_{m-1}\right) \le (2M_\beta^2+4)\exp\left(-\widetilde{C}_{\mathcal{A}} n_{m,a} h_{m,a}^d \epsilon_m^2\right)$$

$$= (4+2M_\beta^2)\exp\left(-\widetilde{C}_{\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2\right).$$

Moreover, marginalizing over the history $\mathcal{F}_{m-1}$ gives

$$\mathbb{P}\left(|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \ge \frac{\epsilon_m}{2} |\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}\right) \le (4+2M_\beta^2)\exp\left(-\widetilde{C}_{\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2\right).$$

Step V: uniform convergence

Since $|\mathcal{R}_{m,a}| \le |\mathcal{X}| = 1$, we can take union bound:

$$\mathbb{P}\left(\sup_{\mathbf{x}\in\mathcal{R}_{m,a}} |\hat{f}_{m,a}(\mathbf{x}) - f_a(\mathbf{x})| \ge \frac{\epsilon_m}{2} |\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a}\right)$$

$$\le \mathbb{P}(|\hat{f}_{m,a}(\mathbf{x}_0) - f_a(\mathbf{x}_0)| \ge \frac{\epsilon_m}{2} |\overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}, N_{m,a} = n_{m,a})$$

$$\le (4+2M_\beta^2)\exp(-\widetilde{C}_{\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2).$$

When $T \ge \exp(\frac{36M_\beta L^2 v_d^2 p_{\max}^2 \widetilde{C}_{\mathcal{A}}(2\beta+d)}{p_0^2 \widetilde{\lambda}_0^2 (2\beta+d+\beta d)})$, it follows that

$$\mathbb{P}(\mathcal{G}_m^C \mid \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_{m-1}) \le A(4+2M_\beta^2)\exp(-\widetilde{C}_{\mathcal{A}} \min_{a\in\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2).$$

Furthermore, under event $\mathcal{M}_m$we have $N_{m,a} \ge T_m$, thus

$$\begin{aligned}\mathbb{P}(\mathcal{G}_m^C \mid \overline{\mathcal{G}}_{m-1}, \overline{\mathcal{M}}_m) &\le A(4+2M_\beta^2)\exp(-\widetilde{C}_{\mathcal{A}} \min_{a\in\mathcal{A}} n_{m,a}^{\frac{2\beta}{2\beta+d}} \epsilon_m^2) \\ &\le A(4+2M_\beta^2)\exp(-\widetilde{C}_{\mathcal{A}} T_m^{\frac{2\beta}{2\beta+d}} \epsilon_m^2) = \frac{A(4+2M_\beta^2)}{T},\end{aligned}$$

where $\epsilon_m = T_m^{-\frac{\beta}{d+2\beta}} \sqrt{\frac{\log T}{\widetilde{C}_{\mathcal{A}}}}$.

Proof of Lemma EC.12. The proof of this lemma is detailed in Rigollet and Zeevi (2010), thus we omit it.

Proof of Lemma EC.13. For any policy π and any $t = 1, \cdots, t_i$ and $j = 1, \cdots, m$, denote the joint distribution of $\{\mathbf{x}_l\}_{l=1}^{t} \cup \{y_{l,a_l}\}_{l=1}^{\Gamma(t)-1}$ as $\mathbb{P}^t_{\pi,\sigma}$, where $\Gamma(t)$ is the batch index for t and $(\mathbf{x}_t, y_{t,1}, y_{t,-1})$ are generated i.i.d from $\mathbb{P}_\sigma$. Drawing upon the definition of the infinite sample rate and the instance constructed above, we can deduce that

$$\sup_{\mathbb{P}\in\mathcal{H}_q} I_{t_i}(\pi) = \sup_{\sigma\in\Sigma_m} I_{t_i}(\pi) = \sup_{\sigma\in\Sigma_m} \sum_{t=1}^{t_i} \mathbb{E}^t_{\pi,\sigma}[\mathbf{1}\{\pi_t(\mathbf{x}_t) \neq \mathrm{sign}(f_1^\sigma(\mathbf{x}_t))\}]$$

$$= \sup_{\sigma\in\Sigma_m} \sum_{j=1}^{m} \sum_{t=1}^{t_i} \mathbb{E}^t_{\pi,\sigma}[\mathbf{1}\{\pi_t(\mathbf{x}_t) \neq \sigma_j, \mathbf{x}_t \in \mathcal{X}_j\}]$$

$$\geq \frac{1}{2^m} \sum_{j=1}^{m} \sum_{t=1}^{t_i} \sum_{\sigma\in\Sigma_m} \mathbb{E}^t_{\pi,\sigma}[\mathbf{1}\{\pi_t(\mathbf{x}_t) \neq \sigma_j, \mathbf{x}_t \in \mathcal{X}_j\}]$$

$$= \frac{1}{2^m} \sum_{j=1}^{m} \sum_{t=1}^{t_i} \sum_{\sigma_{[-j]}\in\Sigma_{m-1}} \sum_{i\in\{-1,1\}} \mathbb{E}^t_{\pi,\sigma^i_{[-j]}}[\mathbf{1}\{\pi_t(\mathbf{x}_t) \neq i, \mathbf{x}_t \in \mathcal{X}_j\}],$$

where $\sigma_{[-j]} = (\sigma_1, \cdots, \sigma_{j-1}, \sigma_{j+1}, \cdots, \sigma_m)$ and $\sigma^i_{[-j]} = (\sigma_1, \cdots, \sigma_{j-1}, i, \sigma_{j+1}, \cdots, \sigma_m)$. Relating the results above to a binary testing error,

$$\sum_{i\in\{-1,1\}} \mathbb{E}^t_{\pi,\sigma^i_{[-j]}}[\mathbf{1}\{\pi_t(\mathbf{x}_t) \neq i, \mathbf{x}_t \in \mathcal{X}_j\}] = \frac{1}{q^d} \sum_{i\in\{-1,1\}} \mathbb{P}^t_{\pi,\sigma^i_{[-j]}}(\pi_t(\mathbf{x}_t) \neq i | \mathbf{x}_t \in \mathcal{X}_j)$$

$$\geq \frac{1}{4q^d} \exp[-\mathcal{K}(\mathbb{P}^t_{\pi,\sigma^{-1}_{[-j]}}, \mathbb{P}^t_{\pi,\sigma^{1}_{[-j]}})],$$

where the second inequality invokes Le Cam's method and $\mathcal{K}(\cdot,\cdot)$ denotes the KL-divergence of two probability distributions (Kullback and Leibler 1951). Under the batch learning setting, the available information is only up to $t_{\Gamma(t)-1}$, thus

$$\begin{aligned}
\mathcal{K}(\mathbb{P}^t_{\pi,\sigma^{-1}_{[-j]}}, \mathbb{P}^t_{\pi,\sigma^{1}_{[-j]}}) = \mathcal{K}(\mathbb{P}^{t_{\Gamma(t)-1}}_{\pi,\sigma^{-1}_{[-j]}}, \mathbb{P}^{t_{\Gamma(t)-1}}_{\pi,\sigma^{1}_{[-j]}}) &\overset{\text{(i)}}{\leq} \frac{1}{\kappa^2} \mathbb{E}_{\pi,\sigma^{-1}_{[-j]}}\Bigg[\sum_{T=1}^{t_{\Gamma(t)-1}} \big(f_{\sigma^{-1}_{[-j]}}(\mathbf{x}_t) - f_{\sigma^{1}_{[-j]}}(\mathbf{x}_t)\big)^2 \mathbf{1}\{\pi_T(\mathbf{x}_t) = 1\}\Bigg] \\
&\overset{\text{(ii)}}{\leq} \frac{4C_\phi^2}{\kappa^2 q^{2\beta}} \mathbb{E}_{\pi,\sigma^{-1}_{[-j]}}\Bigg[\sum_{T=1}^{t_{\Gamma(t)-1}} \mathbf{1}\{\pi_T(\mathbf{x}_t) = 1, \mathbf{x}_t \in \mathcal{X}_j\}\Bigg] \\
&\overset{\text{(iii)}}{=} \frac{4C_\phi^2}{\kappa^2 q^{2\beta+d}} \sum_{T=1}^{t_{\Gamma(t)-1}} \mathbb{P}^t_{\pi,\sigma^{-1}_{[-j]}}\big(\pi_T(\mathbf{x}_t) = 1 \mid \mathbf{x}_t \in \mathcal{X}_j\big) \\
&\overset{\text{(iv)}}{\leq} \frac{4C_\phi^2 t_{\Gamma(t)-1}}{\kappa^2 q^{2\beta+d}} \leq \frac{t_{\Gamma(t)-1}}{\kappa^2 q^{2\beta+d}}.
\end{aligned}$$

Here, step (i) uses the standard decomposition of KL-divergence and the Bernoulli reward structure; step (ii) is due to the definition of f_σ; step (iii) uses $\mathbb{P}(\mathbf{x}_t \in \mathcal{X}_j) = q^{-d}$, and step (iv) arises from $\mathbb{P}^t_{\pi,\sigma^{-1}_{[-j]}}\big(\pi_T(\mathbf{x}_t) =$

$1 \mid \mathbf{x}_t \in \mathcal{X}_j\big) \leq 1$ for any $1 \leq T \leq t_{\Gamma(t)-1}$ and $C_\phi \leq \delta_0 = \frac{1}{2}$. Incorporating the aforementioned results, we arrive that

$$\begin{aligned}
\sup_{\mathbb{P}\in\mathcal{H}_q} I_{t_i}(\pi) &\geq \frac{1}{2^m}\sum_{j=1}^{m}\sum_{t=1}^{t_i}\sum_{\sigma_{[-j]}\in\Sigma_{m-1}}\frac{1}{4q^d}\exp[-\mathcal{K}(\mathbb{P}^t_{\pi,\sigma^{-1}_{[-j]}},\mathbb{P}^t_{\pi,\sigma^{1}_{[-j]}})]\\
&\geq \frac{2^{m-1}}{2^m}\sum_{j=1}^{m}\sum_{t=1}^{t_i}\frac{1}{4q^d}\exp\left(-\frac{t_{\Gamma(t)-1}}{\kappa^2 q^{2\beta+d}}\right)\\
&\geq \frac{1}{8}\sum_{j=1}^{m}\sum_{l=1}^{i}\frac{t_l-t_{l-1}}{q^d}\exp\left(-\frac{t_{l-1}}{\kappa^2 q^{2\beta+d}}\right)\\
&\geq \frac{1}{8}\sum_{j=1}^{m}\sum_{l=1}^{i}\frac{t_l-t_{l-1}}{q^d}\exp\left(-\frac{t_{i-1}}{\kappa^2 q^{2\beta+d}}\right),
\end{aligned}$$

where the last inequality holds since $t_{l-1}\leq t_{i-1}$ for all $1\leq l\leq i$. Recall that $q=q_i=\lceil(\frac{t_{i-1}}{4\kappa^2})^{\frac{1}{2\beta+d}}\rceil$ for $i\geq 2$, $q=1$ for $i=1$ and $m=\lceil q^{d-\alpha\beta}\rceil$. It is possible to further extend the lower bound for some $c^\star>0$,

$$\begin{aligned}
\sup_{\mathbb{P}\in\mathcal{H}_q} I_{t_i}(\pi) &\geq \frac{1}{8}\sum_{j=1}^{m}\sum_{l=1}^{i}\frac{t_l-t_{l-1}}{q^d}\exp\left(-\frac{t_{i-1}}{\kappa^2 q^{2\beta+d}}\right)\\
&\geq c^\star\sum_{j=1}^{q^{d-\alpha\beta}}\sum_{l=1}^{i}\frac{t_l-t_{l-1}}{q^d}\\
&= c^\star\cdot\frac{t_i}{q^{\alpha\beta}} = \begin{cases} c^\star\cdot\frac{t_i}{t_{i-1}^{\frac{\alpha\beta}{2\beta+d}}}, & i>1\\ c^\star t_1, & i=1\end{cases}.
\end{aligned}$$

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